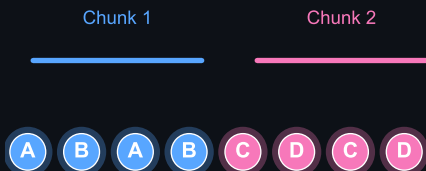


Structured Memory Without Reusable Slots

How RNNs Represent Hierarchical Sequences

Nested Sequences



The Question

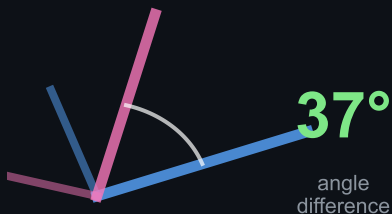
How do neural networks represent hierarchical sequence structure?



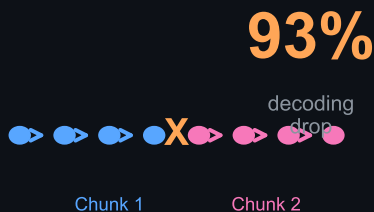
Competing Theories

- Slots**
Reusable scaffolds
- Attractors**
Continuous conveyor
- Buffers**
Structure-specific

Finding 1: Hierarchical Geometry



Finding 2: Boundary Block



Finding 3: No Reusable Slots



Consistent with Memory Buffers (El-Gaby et al.)

RNNs develop structure-specific, conjunctive representations rather than factorized slots

Slots

X

Attractors

X